

e) $\neg(p \rightarrow q) \rightarrow p$
 Suppose for the sake of contradiction, this is not a tautology. $\therefore$ There exists an assignment of truth values to p, q such that $\neg(p \rightarrow q) \rightarrow p$ is False.
 By defn. of conditional, $\neg(p \rightarrow q)$ is True and p is False.
 By defn. of negation, $(p \rightarrow q)$ is False. By defn. of conditional, p is True but q is False. This is a contradiction.
 $\therefore$ It is a tautology

f) $\neg(p \rightarrow q) \rightarrow \neg q$
 Suppose for the sake of contradiction, this is not a tautology. $\therefore$ There exists an assignment of truth values to p, q such that $\neg(p \rightarrow q) \rightarrow \neg q$ is False.
 By defn. of conditional, $\neg(p \rightarrow q)$ is True and $\neg q$ is False
 By defn. of negation, $p \rightarrow q$ is False and q is True
 By defn. of conditional, p is True but q is False. This is a contradiction.
 $\therefore$ It is a tautology

4) Show that each conditional statement in Exercise 12 is a tautology, using the fact that a conditional statement is false exactly when the hypothesis is True and the conclusion is False. (Do not use truth tables)

a) $[\neg p \wedge (p \vee q)] \rightarrow q$
 Suppose for the sake of contradiction, this is not a tautology. $\therefore$ There exists an assignment of truth values to p, q such that $[\neg p \wedge (p \vee q)] \rightarrow q$ is False.
 By defn. of conditional, $\neg p \wedge (p \vee q)$ is True, but q is False.
 By defn. of AND, $\neg p$ is True and $(p \vee q)$ is True.
 By defn. of negation, p is False. By defn. of OR, since p is False, q is True.
 But this is a contradiction. $\therefore$ It is a tautology

b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
 Suppose for the sake of contradiction, this is not a tautology. $\therefore$ There exists an assignment of truth values to p, q, r such that $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is False.